



IAWN Update

Lindley Johnson

Planetary Defense Officer Emeritus

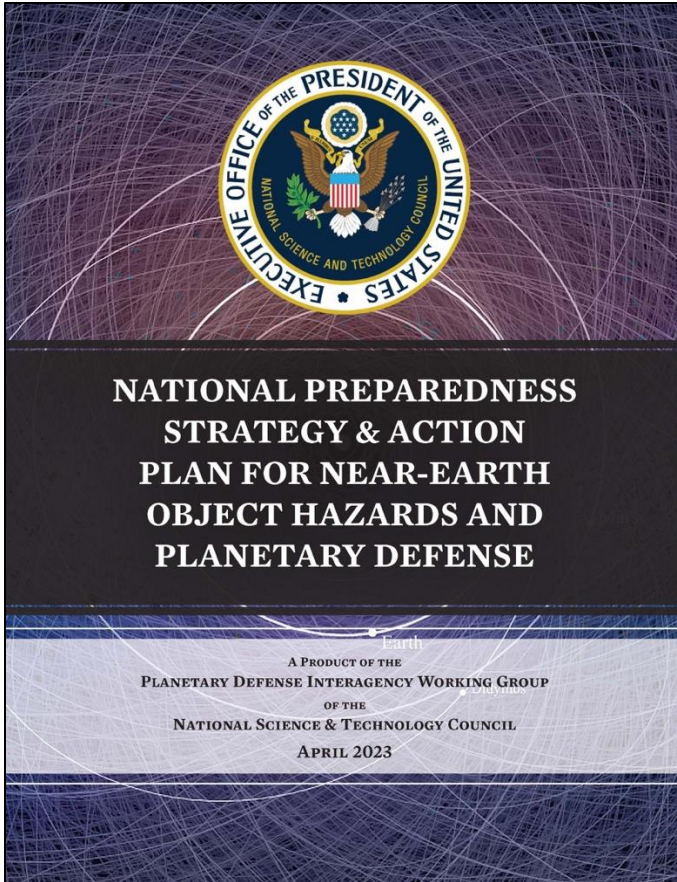
Kelly Fast

Planetary Defense Officer (Acting) & NEO Observations Program Manager

September 17, 2024



Planetary Defense for the Next Decade (2023 - 2033)



National Plan Goals (Critical Focus Goals in Red)

1. Enhance NEO detection, tracking, and characterization capabilities
2. Improve NEO modeling, prediction, and information integration
3. Develop technologies for NEO reconnaissance, deflection, and disruption missions
4. Increase international cooperation on NEO preparedness
5. Strengthen and routinely exercise NEO impact emergency procedures and action protocols
6. Improve U.S. management of planetary defense through enhanced interagency collaboration



ASSESS

Determine NEO population survey completeness and hazard from NEOs that pose the highest risk



SEARCH, DETECT & TRACK

Find the natural near-Earth objects – asteroids and comets – and track to determine those whose orbits create an impact hazard to Earth

PLANETARY DEFENSE

MITIGATE

Demonstrate technologies and techniques to divert or disrupt asteroids in space or inform emergency response activities on the ground

PLAN & COORDINATE

Work with the U.S. interagency and international collaborations on effective actions for impact threat response

CHARACTERIZE

Determine physical characteristics of NEOs (size, shape, composition, rotation) to understand their natural state



Strengthening the International Response to Complex Global Shocks - An Emergency Platform

FACTORS THAT CONTRIBUTE TO THE DECISION TO CONVENE AN EMERGENCY PLATFORM

The Secretary-General would decide when to convene an Emergency Platform in response to a complex global shock.

In advance of the decision, the Secretary-General would consult with:

- The President of the General Assembly;
- The President of the Security Council (as appropriate);
- Relevant national authorities and/or regional organizations;
- Relevant United Nations entities, specialized agencies, international financial institutions, and other multilateral institutions and agencies that have been mandated by Member States to respond to sector-specific crises.

EXERCISE EXERCISE EXERCISE

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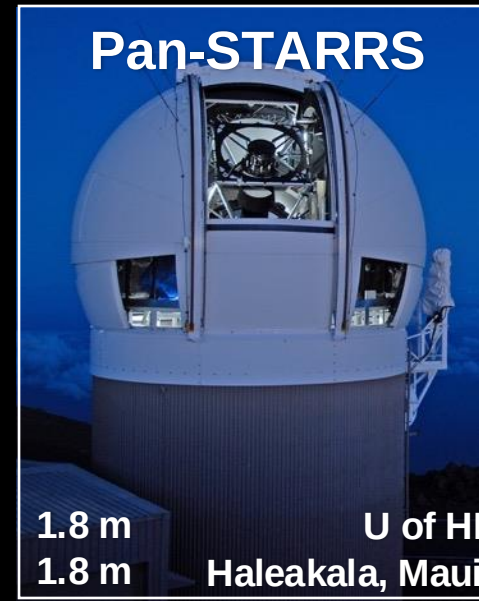
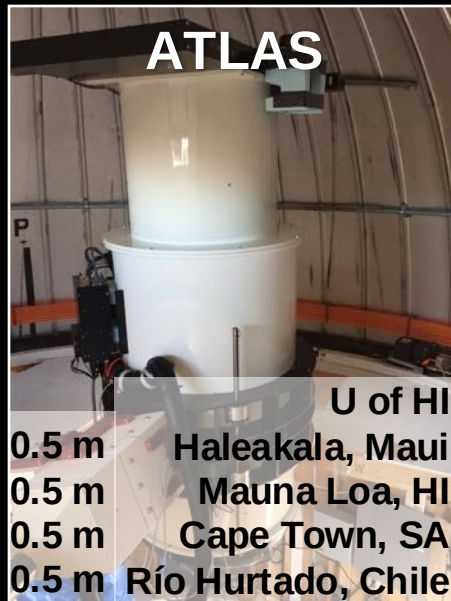
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EXERCISE EXERCISE EXERCISE

Planetary Defense TTX #5 After Action Report:
<https://cneos.jpl.nasa.gov/pd/cs/ttx24/final.html>

NASA's Current NEO Program Search/Survey Systems



NEOWISE Decommissioned Aug. 8, 2024

- NEOWISE relayed its final data to Earth before the project team at NASA JPL sent a command that turned off its transmitter
- This concludes more than 10 years of its planetary defense mission to search for potentially hazardous NEOs
- The spacecraft is expected to reenter Earth's atmosphere before the end of the year
- NEOWISE helped to inform development of NASA's NEO Surveyor, which is expected to launch in 2027

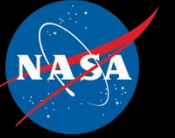


NEO Surveyor

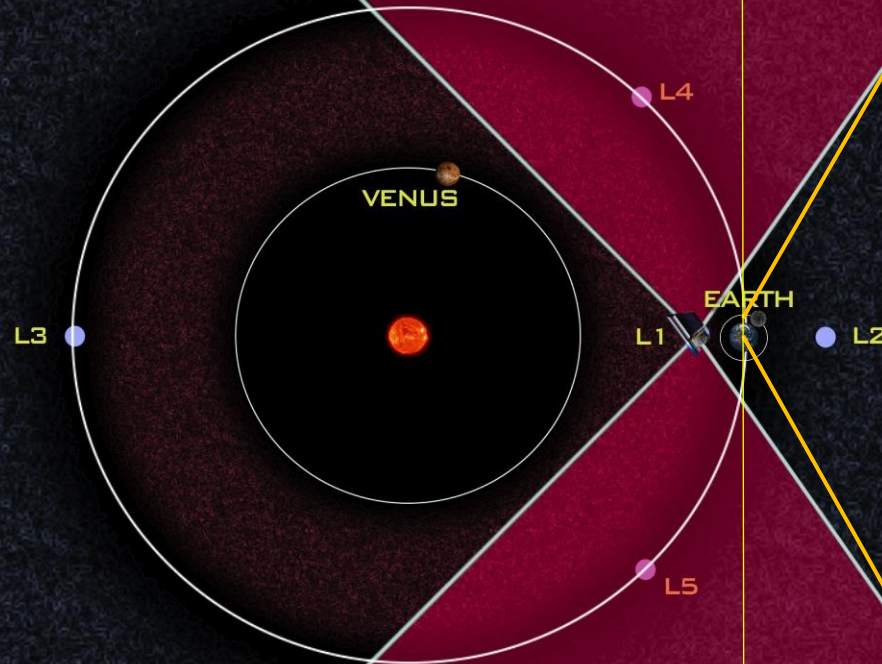


NEO Surveyor
field-of-regard

NEOWISE
field-of-regard



- Space-based infra-red telescope
- Objectives:
 - Find 65% of Potentially Hazardous Asteroids (PHAs) >140 m in 5 years (>90% in 10 years)
 - Better estimate object sizes



Area at Opposition seen
by ground-based assets

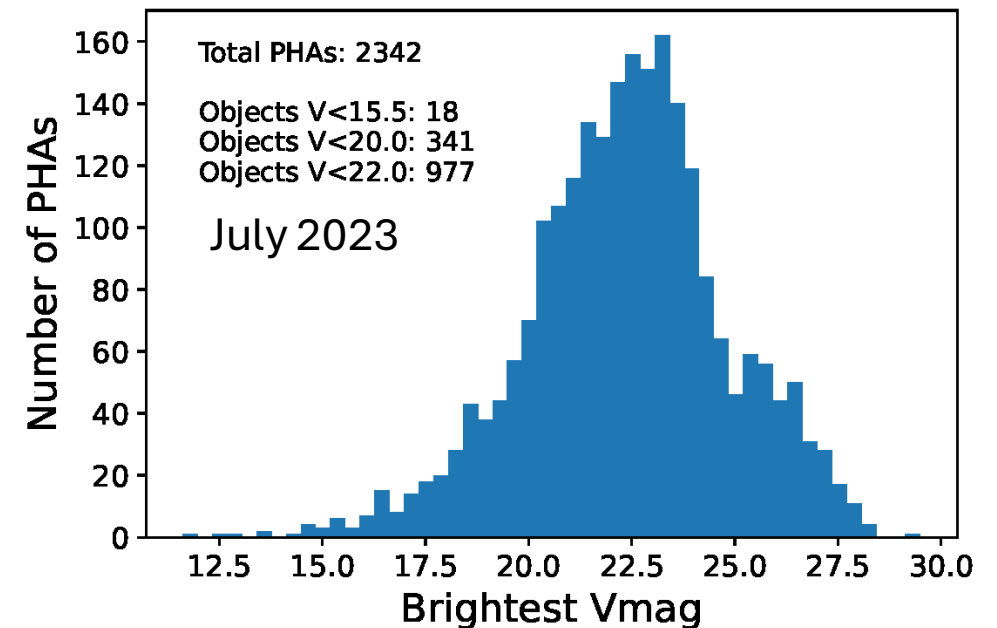
- **KDP-C Confirmation Nov. 29 2022 – mission now in Phase C**
- **Project progressing towards Launch Readiness in Fall 2027**

SPECTRE on NASA IRTF

- NASA's Infrared Telescope Facility (IRTF) is a 3.2m infrared-optimized telescope on Mauna Kea, Hawaii (with remote observing capability) open to the entire astronomical community via peer review: <https://irtfweb.ifa.hawaii.edu/>
- SPECTRE is a new instrument being built to characterize small bodies (and astrophysics transients) with the NASA IRTF
- We can rapidly characterize NEOs using a single instrument over multiple wavelengths (visible, near-IR and thermal-IR)
- It is complimentary to SpeX but enables us to characterize fainter NEOs
- Ideal for NEOs discovered by surveys and by next generation surveys such as NEO Surveyor and Vera Rubin Observatory
- SPECTRE will enable us to constrain taxonomy, composition and albedo/diameter of an NEO during IAWN campaigns.

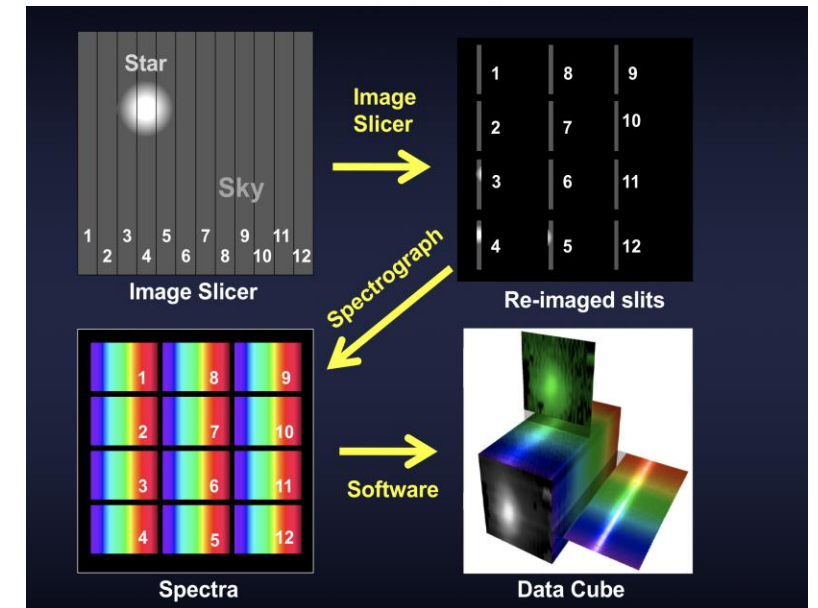
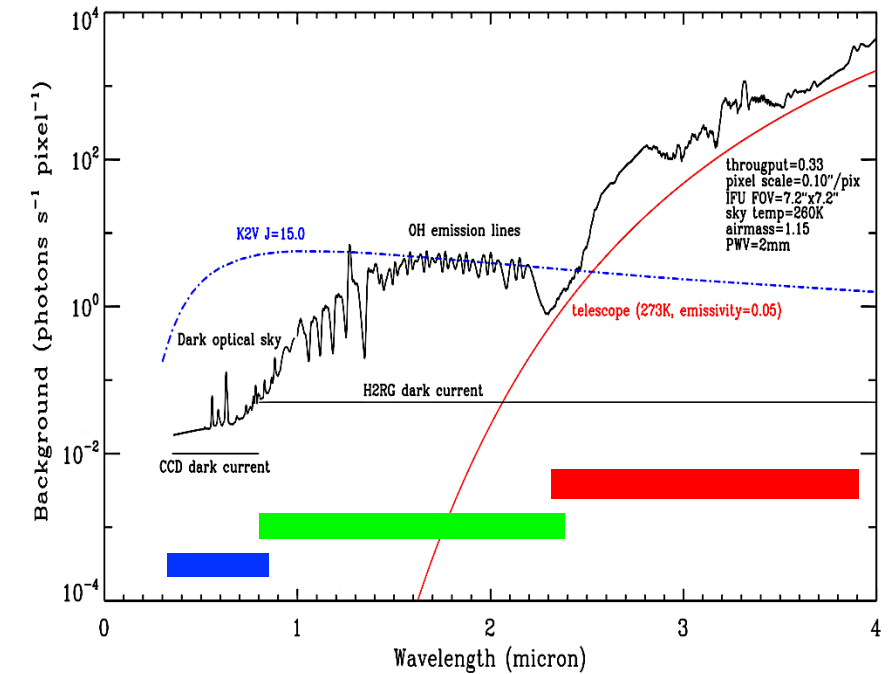
Limiting magnitude for SPECTRE
SNR~5 Exposure time 1000 seconds

Wavelength (μm)	Visual Magnitude
0.56	22.7
0.75	22.9
1.33	20.4
3.0	15.5

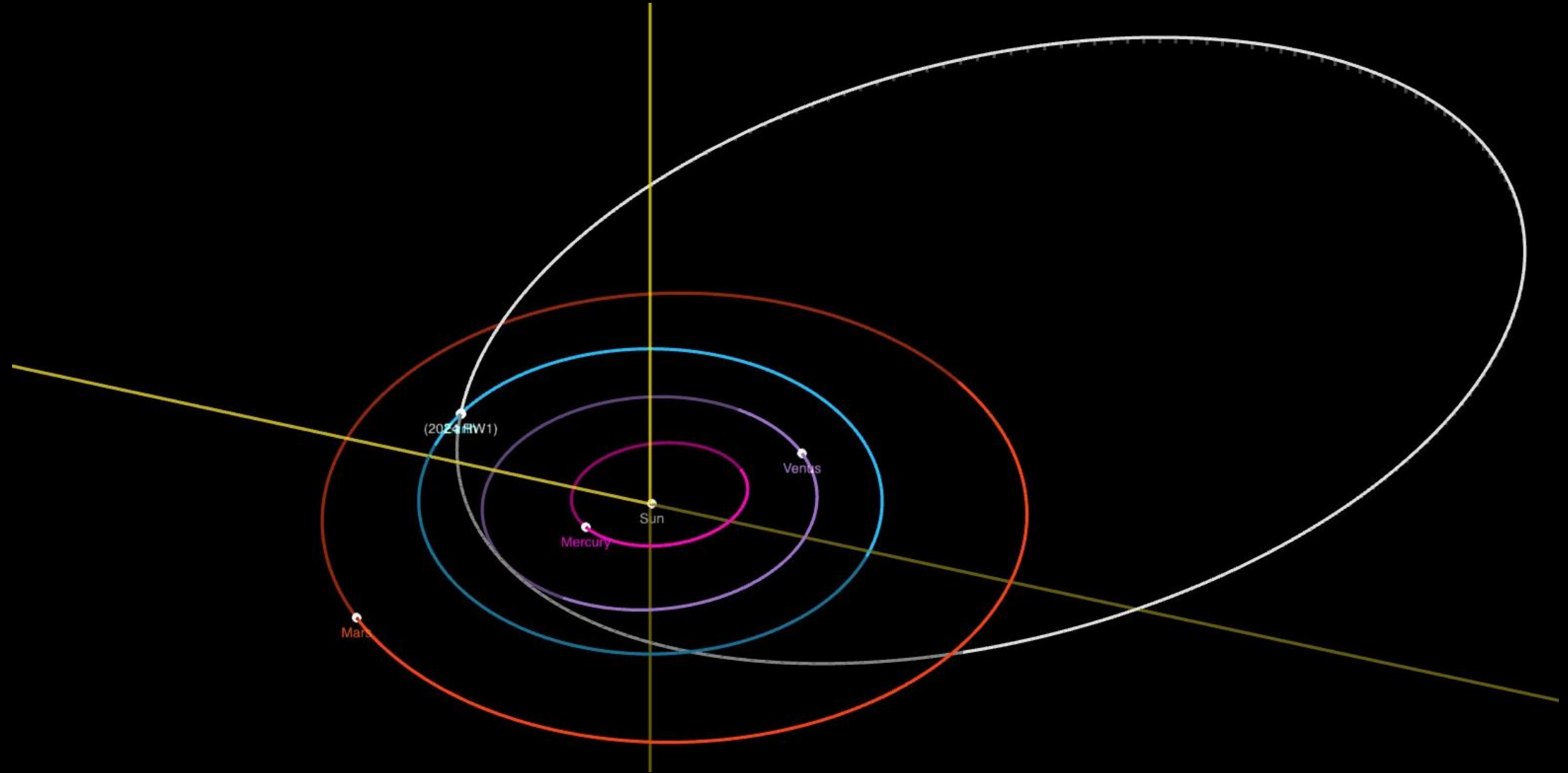


SPECTRE: one-shot 0.4-4.2 μm IFU **$R=150$ SPECTROGRAPH FOR IRTF**

- Three channels: **visible**, **NIR**, **MIR**
- (**$2k^2$ CCD**, **H2RG/2.5 μm** , **H2RG/5.3 μm**)
- 3x3 arc-min CCD for A&G and fast guiding
- Compact and rigid, no moving parts
- Image slicer IFU with 7" x 7" FOV
 - Point and shoot
 - No slit losses, high throughput (70%)
 - Absolute spectro-photometry
 - Accurate spectral slopes
 - Removes ambiguity of rotation and phase angle
 - One mag + sensitivity gain + multiplex gain
- First light 2027 (estimated)



Impact of small asteroid 2024 RW1 on September 4, 2024



* *The impact was off the east coast of the northern Philippines on 9/4/24 at 12:39pm EDT*

Impact of small asteroid 2024 RW1 on September 4, 2024

 **Catalina Sky Survey** @catalina_sky · Sep 4
BIG NEWS! CSS observer Jacqueline Fazekas has just discovered a small meteoroid that will harmlessly impact with Earth's atmosphere in about 8 hours over the Western Pacific. The object (called CAQTDL2) is small, a few feet in diameter. Attached are the discovery images.



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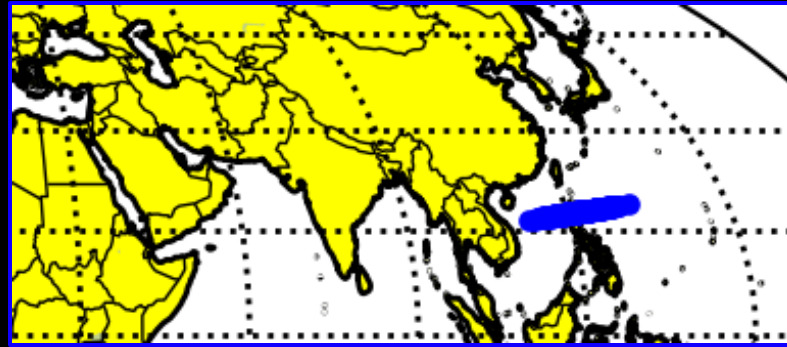
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759

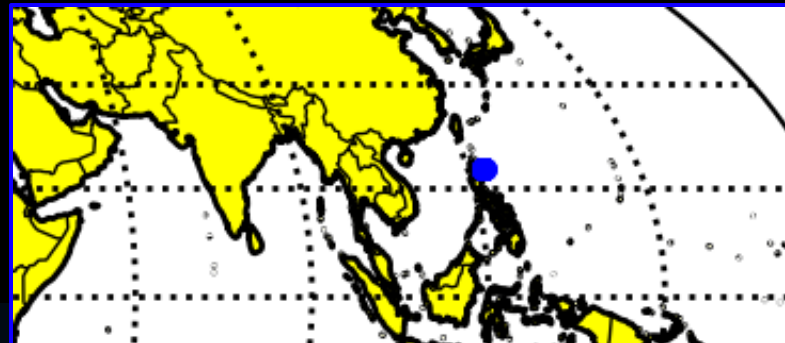
106K



NASA JPL's Center for Near-Earth Object Studies (CNEOS) issued an alert 10 hours before impact based on observations sent to the Minor Planet Center by Jacqueline Fazekas at the NASA-funded Catalina Sky Survey



Follow-up from observatories across the globe allowed CNEOS to quickly refine the impact prediction to off the east coast of the northern Philippines. ESA NEOCC made a similar prediction.

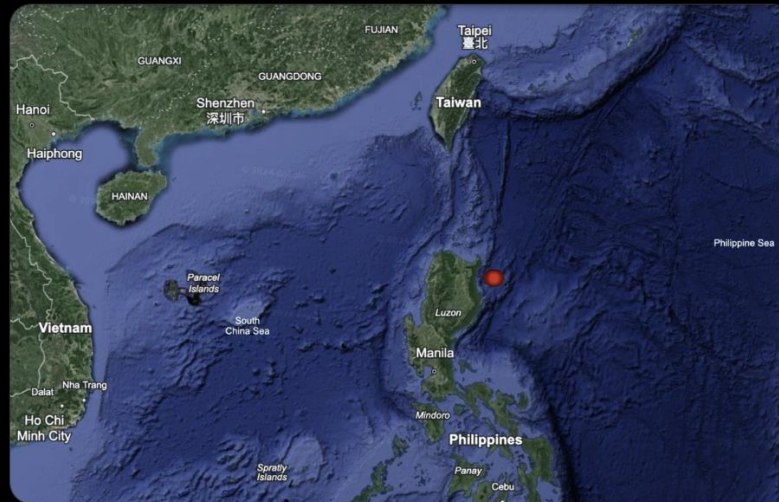


Impact of small asteroid 2024 RW1 on September 4, 2024

- The asteroid posed no threat since it was so small at ~1 meter in size
- It was an excellent demonstration of planetary defense capabilities: find, track and accurately predict impact location
- NASA (@AsteroidWatch) and ESA (@ESA) notified the public on social media channels
- Many watched the impact as it happened and reshared video on social media



.@NASA CNEOS & ESA NEOCC predict that a very small (~1 meter) #asteroid will harmlessly impact Earth's atmosphere in a few hours at ~16:40 UTC (~12:40 pm EDT) and create a #fireball off the east coast of the northern Philippines. It was first observed in space by @catalina_sky.



8:48 AM · Sep 4, 2024 · 141K Views



🌟🔥 Woah! Newly discovered 1-1.5m asteroid 2024 RW1 (spotted by near-Earth asteroid hunters in the US) burns up above the Philippines just hours after it was first seen.

This asteroid wasn't a risk. But if it was, a duck-and-cover warning was possible.

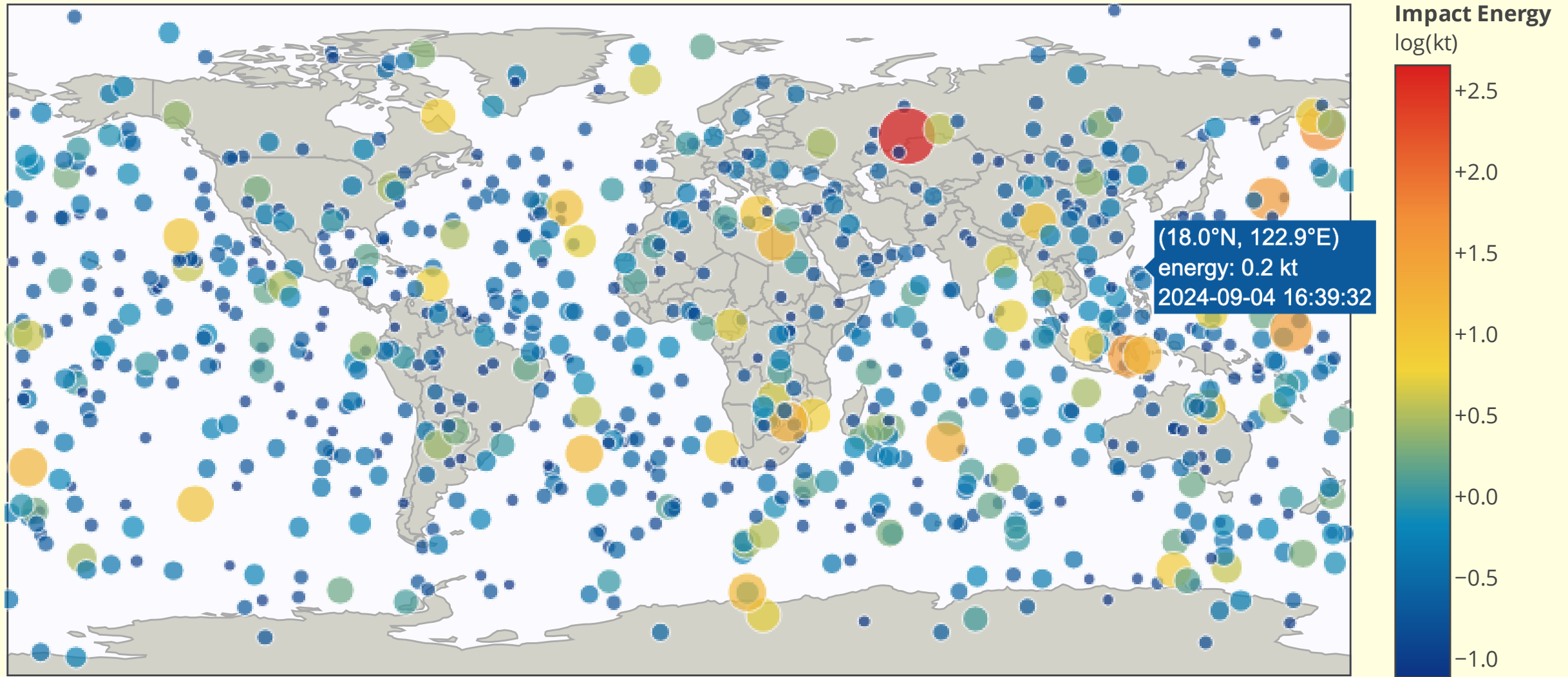


From ScienceKonek

<https://twitter.com/SquigglyVolcano/status/1831383862435578335>

Fireballs Reported by US Government Sensors

(1988-Apr-15 to 2024-Sep-04)

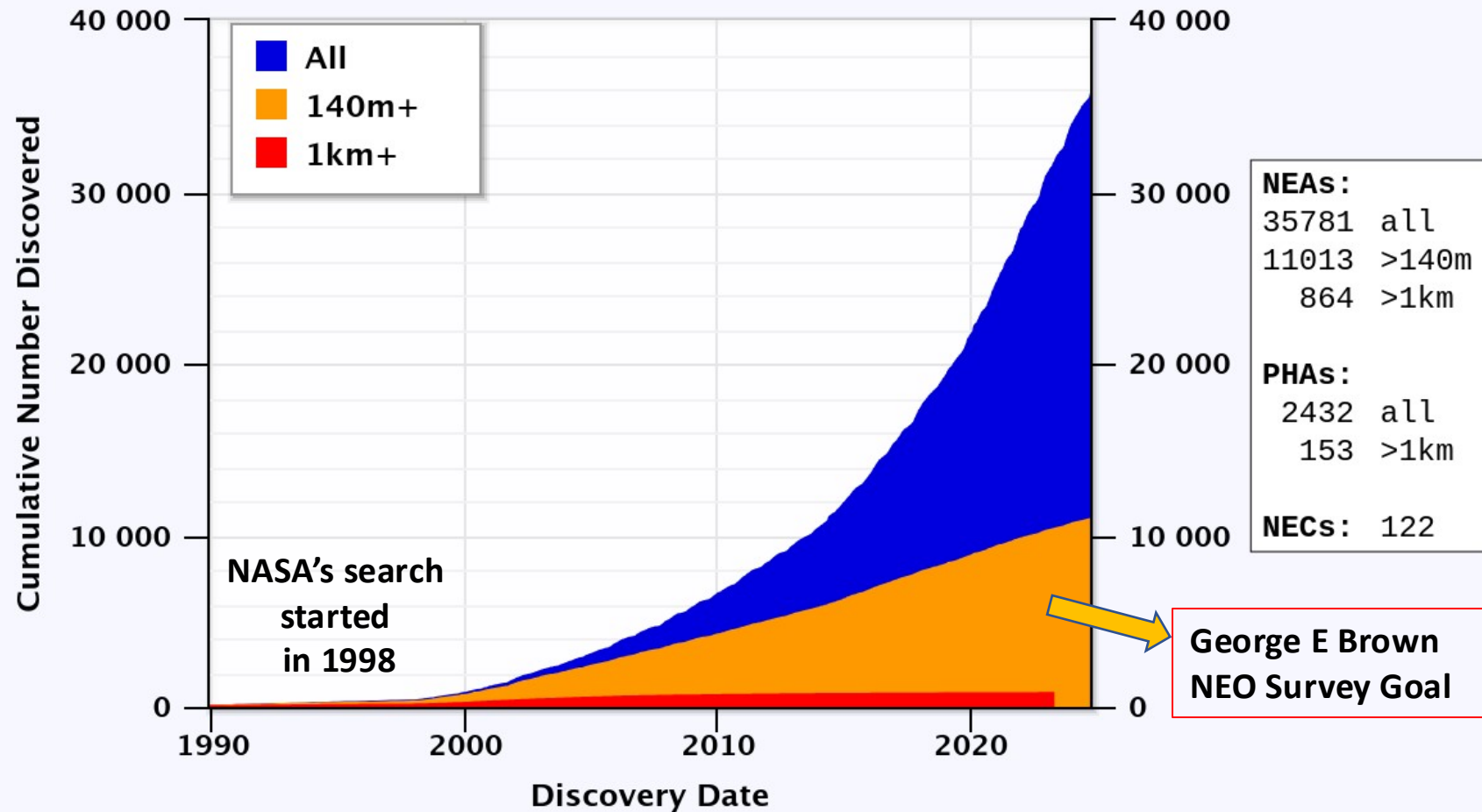


<https://cneos.jpl.nasa.gov/fireballs/>

Alan B. Chamberlin (JPL/Caltech)

Near-Earth Asteroids Discovered

Most recent discovery: 2024-Sep-14



<https://cneos.jpl.nasa.gov/stats/>

Alan Chamberlin (JPL/Caltech)

*Potentially Hazardous Asteroids come within 7.5 million km of Earth orbit

Alan Chamberlin (JPL/Caltech)

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